



INNOMATICS[®]
RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Hospital Management Database

SQL Analytics Project

Turning hospital operations into an analyzable data model

PROJECT PURPOSE AND SCOPE

The project creates **hospital_db** and organizes hospital operations into five connected entities. Its design supports demand analysis, treatment-performance monitoring, workforce analysis, and room-and-equipment utilization.



Patient lifecycle

Profiles, segments, registration



Appointment demand

Services, priorities, channels



Treatment execution

Status, duration, waiting time



Workforce performance

Doctors, workload, outcomes



Room and equipment utilization

Capacity, availability, room type, and equipment context for treatment activity



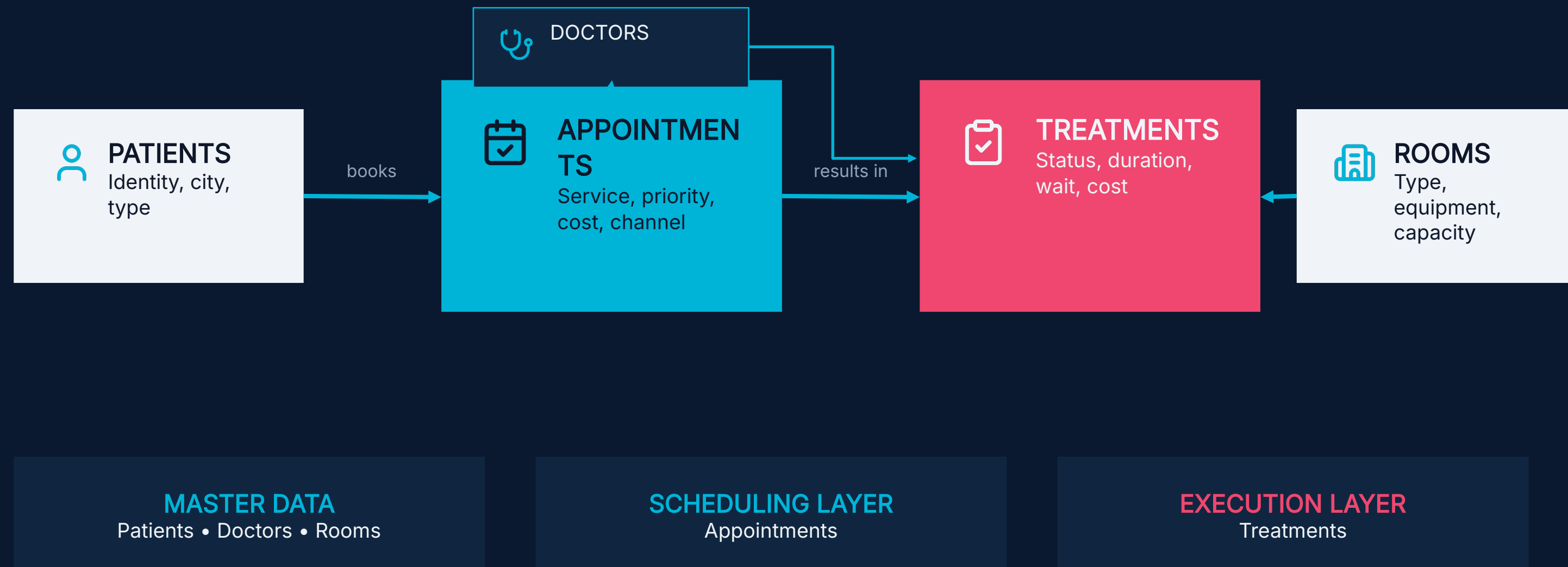
Important framing

The supplied file includes schema definitions and analytical queries, but no INSERT statements or query outputs. This deck therefore presents intended analysis—not fabricated results.

Five core tables form one operational workflow

DATABASE ARCHITECTURE AT A GLANCE

Master data establishes the context; transaction tables capture scheduled and delivered care.



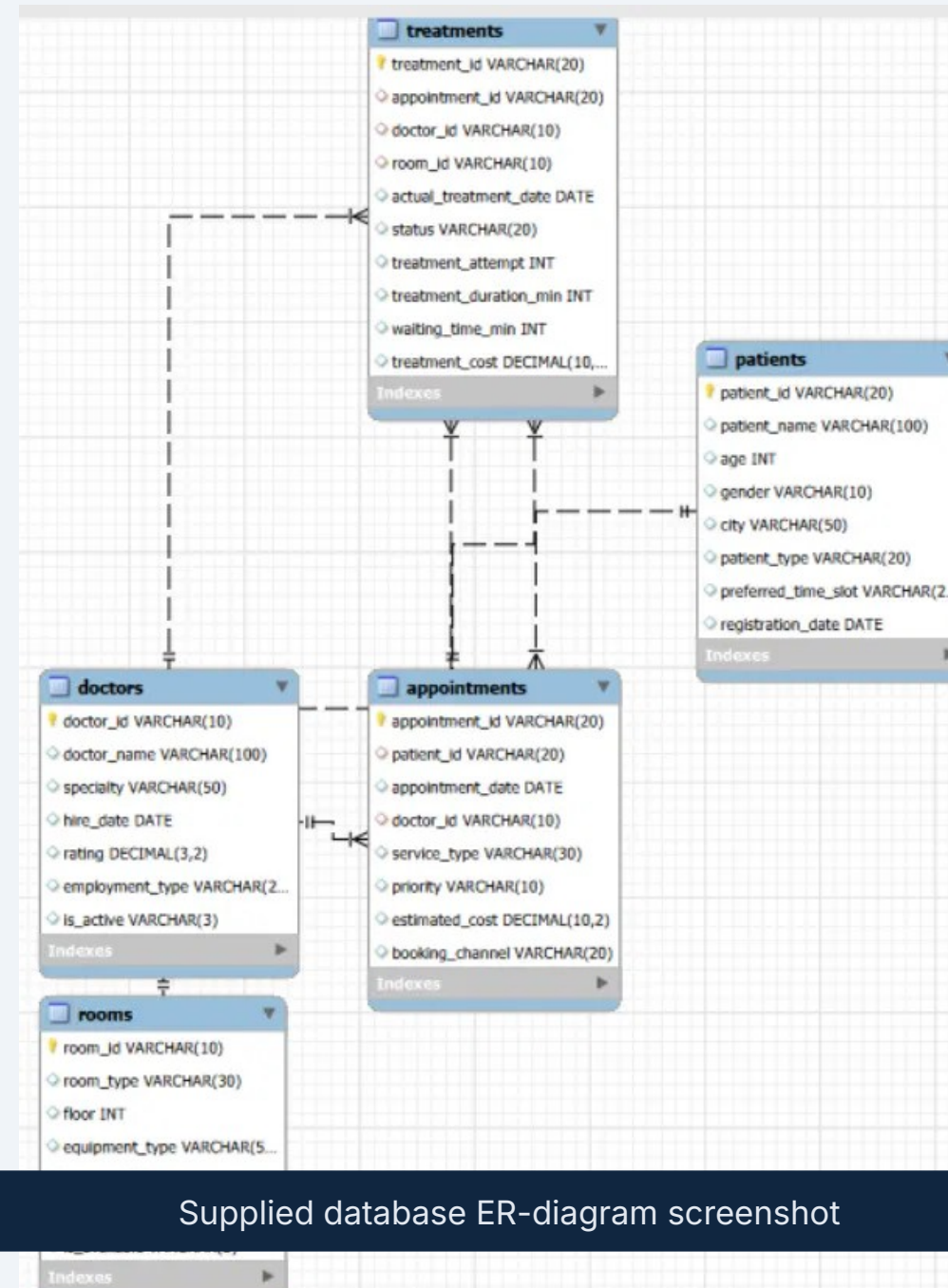
ER diagram: traceability from patient booking to treatment

VISUAL ENTITY-RELATIONSHIP MODEL

How the model works

- **patients** connect to appointments through **patient_id**.
- **doctors** link to both scheduled appointments and delivered treatments.
- **treatments** record status, time, attempts, and cost against appointments.
- **rooms** provide capacity and

The diagram supports a complete journey: **patient** → **appointment** → **treatment** → **outcome**.



Supplied database ER-diagram screenshot

Reading guide

- Primary key
- Foreign-key relationship
- Entity table

Each table has a focused analytical responsibility

TABLE RESPONSIBILITIES AND IMPORTANT ATTRIBUTES

Table	Analytical role	Primary key	Representative attributes
patients	Profile and segmentation	patient_id	age, gender, city, patient_type, preferred_time_slot, registration_date
doctors	Provider directory and quality context	doctor_id	specialty, hire_date, rating, employment_type, is_active
rooms	Physical-resource inventory	room_id	room_type, floor, equipment_type, capacity, last_maintenance_date, is_available
appointments	Demand and booking behavior	appointment_id	appointment_date, service_type, priority, estimated_cost, booking_channel
treatments	Actual service delivery and outcome	treatment_id	status, treatment_attempt, treatment_duration_min, waiting_time_min, treatment_cost

Master data supplies the **who, where, and with what resources**; transaction data supplies the **when, what happened, and how it performed**.



A solid analytical base—with clear production hardening steps

DATA QUALITY AND DESIGN CONSIDERATIONS

What the schema already enables

- ✓ Primary keys are present on every core table.
- ✓ Foreign keys create traceability across patient, doctor, room, appointment, and treatment records.
- ✓ Dates, costs, durations, and counts use suitable analytical data types.
- ✓ The structure supports temporal, operational, utilization, and outcome analysis.

Recommended production safeguards

- Use boolean or controlled lookup values instead of free-text Yes/No flags.
- Add NOT NULL and CHECK constraints for critical fields and valid ranges.
- Validate age, rating, capacity, cost, and duration values.
- Index foreign keys and frequently filtered date, city, status, and service fields.
- Confirm whether appointments can have multiple treatment records by design.

Analysis-ready: schema plus query logic → **Production-ready:** validated data, controls, indexes, governance, and privacy practices

Establishing the operational baseline

BASIC EXPLORATION QUERIES



Tiles show **planned outputs**; no data rows or result values are supplied.

Total patients

COUNT(*) FROM patients

Total appointments

COUNT(*) FROM appointments

Total treatments

COUNT(*) FROM treatments

Service types

DISTINCT service_type

Active doctors

COUNT WHERE is_active = Yes

Room mix

COUNT BY room_type

Estimated value

SUM(estimated_cost)

Average duration

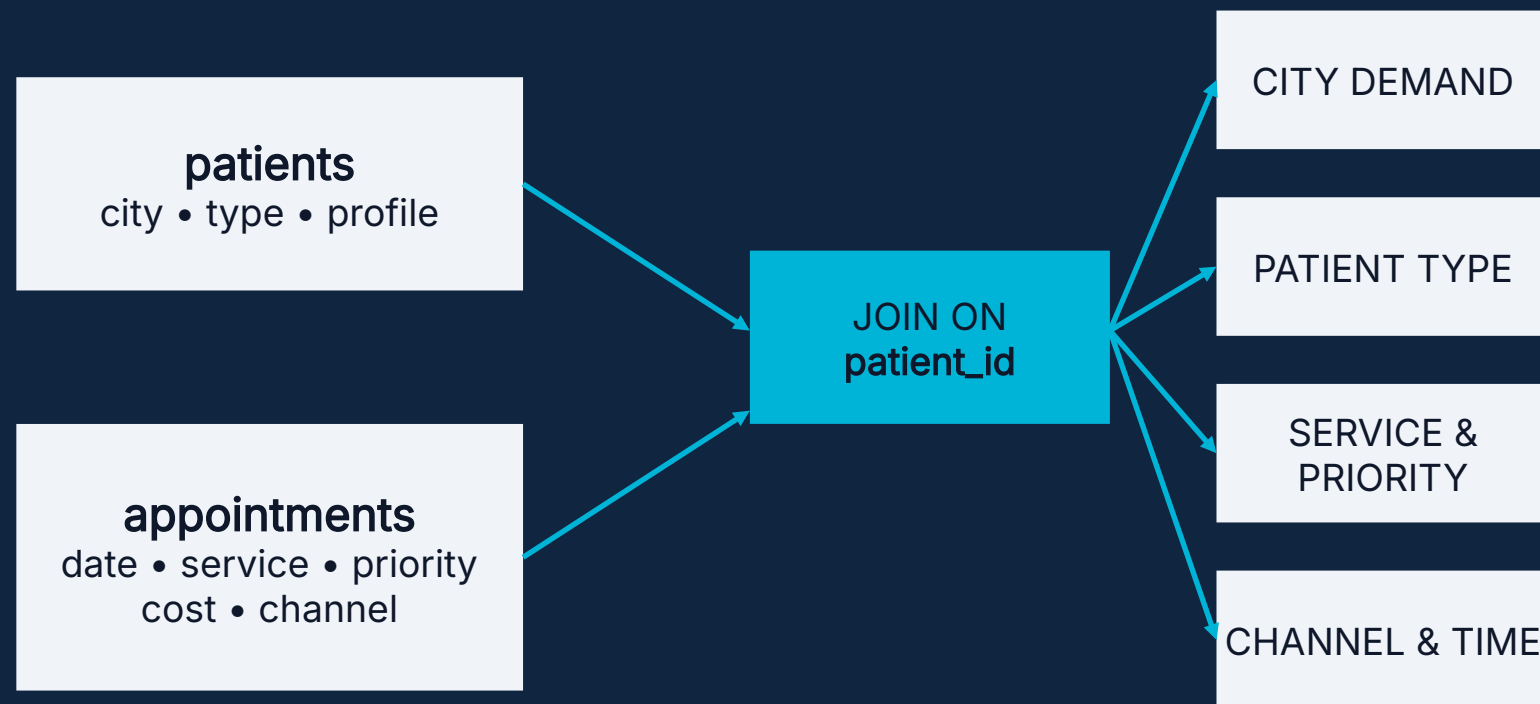
AVG(treatment_duration_min)



Understand who is booking, where demand occurs, and how value is distributed

DEMAND AND PATIENT ACTIVITY ANALYSIS

Query lineage: patients + appointments



Management questions the SQL can answer

- Which cities contribute the greatest appointment volume?
- Which services and priorities drive demand?
- How does booking activity evolve by date and month?
- Which booking channels contribute most appointments?
- How do General, Corporate, and Insurance patients differ in activity and estimated value?

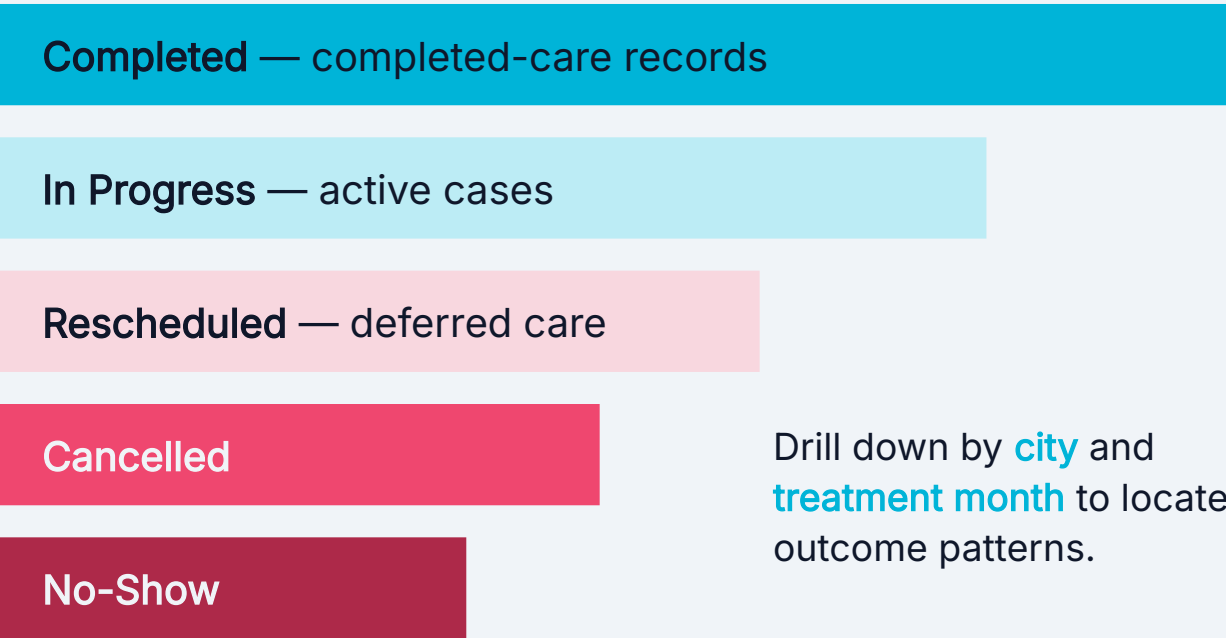
The project measures **estimated appointment value**, not realized treatment revenue.

Move from scheduled demand to delivered care

TREATMENT OUTCOMES AND PATIENT EXPERIENCE

Outcome monitoring

Query: COUNT(*) grouped by treatment status



Drill down by **city** and **treatment month** to locate outcome patterns.

Patient-experience monitoring



Together, status, duration, and wait-time queries can reveal **access bottlenecks**, **non-completion hotspots**, and differences between demand that was scheduled and care that was delivered.

Appointment → Treatment record → Duration + wait → Outcome status



Link operational throughput to people and physical resources

DOCTOR, ROOM, AND EQUIPMENT PERFORMANCE

Provider lens

- Treatment volume by doctor
- Outcome distribution by doctor
- Average, minimum, and maximum treatment duration
- Context fields: specialty, rating, employment type, active status

JOIN doctors → treatments

Resource lens

- Treatment volume by room type and equipment type
- Room-level average duration and wait time
- Average treatment cost by room
- Context fields: capacity, availability, floor, maintenance date

JOIN rooms → treatments

Interpretation lens

Doctor ratings and throughput should be interpreted alongside **case mix**, specialty, workload, service type, and outcome definitions. A single score should never stand alone as a quality conclusion.

CONTEXT BEFORE COMPARISON

Use the model to compare **throughput**, **experience**, and **resource use**—then investigate the drivers behind any differences.

Find friction points that require operational attention

EXCEPTION ANALYSIS AND MANAGEMENT QUESTIONS

Exception signal	SQL logic in project	Management lens
Repeated attempts	MAX(treatment_attempt) > 1 by appointment	Which cases require additional intervention or review?
Longer waits	Compare average waiting time for single vs. multiple attempts	Are repeat attempts associated with service friction?
City non-completion	Count Cancelled, No-Show, and Rescheduled statuses by city	Where may access, scheduling, or communication barriers exist?
Service non-completion	Count exception statuses by service_type	Which services need capacity, process, or booking review?



Ask before acting

- Are exceptions concentrated in a few locations or broadly distributed?
- Are they linked to priority, service type, room use, or booking channel?
- Do waits rise when treatments require multiple attempts?
- Which follow-up data would confirm the root cause?

The supplied SQL defines an **exception-detection framework**; actual priority areas can only be identified after data is loaded and results are reviewed.



THANK
YOU

